Choose your robot's next job

Name: ______________________________

Date: ______________________ **Period:** ______________________

Choose one of five builds, test it and explain how you improved it.

Choose ONE build

1. Parcel delivery robot - full module: 5-8

Carry a paper parcel to a red-line delivery space.

2. Drawing robot - full module: 9-12

Draw a short line on a protected paper surface.

3. Paper sweeper - full module: 13-16

Push three light paper pieces into a collection area.

4. Ball-pushing robot - full module: 17-20

Guide one light paper ball toward a goal.

5. Rescue carrier - full module: 21-24

Tow a light paper figure on a small rescue sled.

Your teacher will give you the pages for one build. Class 1: choose and build. Class 2: test, improve and share. The page numbers above refer to the complete module workbook.

Plan before you build

Our chosen job:

Why we chose it:

LOOK AT THE IDEA

BUILD → **TEST** → **IMPROVE**

Keep your notes. Change one thing. Try again.

Words to know

Attachment: A part you add to the robot for a new job.

Evidence: Measurements and notes that show what happened.

Handover: Directions that help someone else use your robot.

Draw the attachment and label where it joins the base.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Test and prepare to share

Name: ______________________

Date: ______________________ **Period:** ______________________

Use your choice pages to record three runs, change one thing, and test three more times. Keep results from runs that did not work.

Write your handover directions

Set up the test area:

____

____

Start the program:

____

____

Stop the robot:

____

____

Return to the start safely:

____

____

Show what you learned

Give a two-minute demonstration. If it fails, stop safely, explain what happened and show your earlier results.

Our robot's job and how the code works:

A change we made, with a result from before and after:

My part in the team's work:

Another team's feedback and one next test:

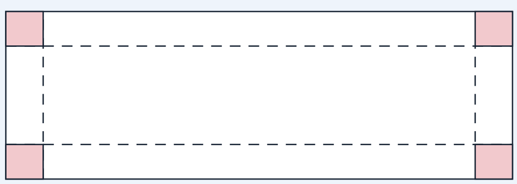
Choice 1: Parcel delivery robot

Name: ______________________

Date: ______________ **Period:** ______________

Carry a paper parcel to a red-line delivery space.

LOOK AT THE IDEA



Cut out the four shaded corners.

Fold on the dashed lines.

Tape corners to make a low tray.

Gather your materials

- The shared checked base
- Thin card about 12 by 8 cm
- One paper parcel about 4 by 3 by 2 cm
- Ruler, scissors and removable tape

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

Build your attachment

1. Measure the clear mounting space on the base. Cut the card to fit it. Keep at least 1 cm of room between the attachment and any wheel. Make it smaller if needed.

2. Draw a fold line 1 cm inside each edge. Cut out the four corner squares. Fold all four walls up and tape the corners to make a shallow tray.

3. Fold a light paper parcel. Place it in the center of the tray. Add a small folded-paper stop at each end if it slides. Do not add heavy contents.

4. Turn the robot off. Use the teacher-approved mounting points and removable connectors to hold the tray flat. Keep buttons and the sensor clear.

5. Raise the wheels and check for rubbing. Then place the robot on the white lane from lesson 6. Keep the broad red strip and empty space beyond it.

6. Run the code below three times. If the parcel falls, change one tray feature and repeat. Unload the parcel by hand only after the robot has stopped.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Program and first tests

Name: ______________________________

Date: ____________________ **Period:** ____________________

1. Use lesson 6’s tested color-stop program without removing its limits.

2. Start stopped; set keepGoing to Yes. Repeat at most 20 times. Only while keepGoing is Yes, read the color: white means a slow 0.1-second move then Stop; any other reading means Stop and set keepGoing to No.

3. Stop and end after the repeats. Red means delivery; unknown means check the setup.

keepGoing remembers whether this test may continue. Yes allows a check; No keeps the robot stopped.

My prediction:

Run	Parcel stayed on?	Inside space?	Hit nothing?
1			
2			
3			

Improve your chosen build

What counts as success?

☐ The parcel stays in the tray.

☐ The whole robot stops inside the delivery space.

☐ The robot hits no object or person.

Change one thing. What did you change?

Run	Parcel stayed on?	Inside space?	Hit nothing?
1			
2			
3			

Which tray feature helped the parcel stay on?

Which result supports your answer?

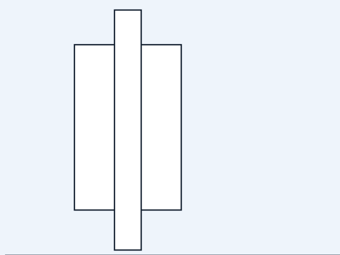
Choice 2: Drawing robot

Name: ____________________

Date: ____________________ **Period:** ____________________

Draw a short line on a protected paper surface.

LOOK AT THE IDEA



Card sleeve

Marker tip just touches the paper.

Wheels stay on the surface.

Gather your materials

- The shared checked base
- One washable felt-tip marker
- A card strip about 12 by 3 cm and removable tape
- A large sheet of paper over a protective mat

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

Build your attachment

1. Cover a large floor area with a protective mat and paper. Tape the paper edges flat. Ask the teacher to approve the marker and surface.
2. Wrap the card strip loosely around the marker to make a sleeve. Tape the card to itself. The marker must slide up and down inside it.
3. Turn the robot off. Fix the sleeve upright at a teacher-approved point behind the wheels. Keep it away from the sensor and moving parts.
4. Slide the uncapped marker down until its tip just touches the paper. Secure the marker in the sleeve with a small removable tape tab. It must not lift the wheels off the paper.
5. Raise the wheels and check the short program. Put the robot back on the protected paper with enough space for the whole run.
6. Draw three short lines. Stop and turn off the robot before lifting or adjusting the marker. If a line has gaps, lower the tip a little; if the robot struggles, raise it.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Program and first tests

Name: ______________________________

Date: ____________________ **Period:** ____________________

1. Start with movement stopped. Use a checked low speed, such as 20 percent.
2. Drive forward for 0.8 seconds, then stop and end. Shorten the time if the paper is too small.
3. No automatic repeat. Lift the marker and return the robot by hand between runs with the power off.

My prediction:

____

Run	Visible line?	Stayed on paper?	Stopped by itself?
1			
2			
3			

Improve your chosen build

What counts as success?

- ☐ A visible line is drawn on the paper.
- ☐ The marker stays in its holder and all marks stay on protected paper.
- ☐ The robot stops by itself at the programmed time.

Change one thing. What did you change?

Run	Visible line?	Stayed on paper?	Stopped by itself?
1			
2			
3			

How did marker height change the line or the robot’s movement?

Which result supports your answer?

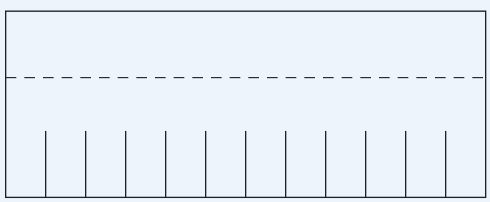
Choice 3: Paper sweeper

Name: ______________________________

Date: ____________________ **Period:** ____________________

Push three light paper pieces into a collection area.

LOOK AT THE IDEA



Fold the top edge back.

Cut soft bristles below the fold.

Keep the wheels clear.

Gather your materials

- The shared checked base
- A thin card strip about 12 by 5 cm
- Three tissue-paper squares, each about 2 cm wide
- Removable tape, ruler and scissors

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

Build your attachment

1. Cut a card strip as wide as the robot, but no wider than the clear test lane. Fold its top 2 cm backward to make a mounting edge.
2. Cut short vertical slits about 1 cm apart into the lower 2 cm. These make flexible paper bristles. Round sharp card corners with scissors.
3. Turn the robot off. Attach the folded top edge to approved fixed front mounting points. The flexible ends should just brush the floor without lifting any wheel.
4. Use a clear floor lane. Mark a collection box farther along it. Place three tissue squares in a row across the starting sweep area. Keep them away from wheel paths.
5. Try the short program below. Check that the paper bristles can bend. Stop at once if a piece catches in a wheel; turn the power off before removing it.
6. Count the pieces that reach the box. Repeat three times with the same starting positions. Adjust the bristle height once and compare.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Program and first tests

Name: ______________________________

Date: ____________________ **Period:** ____________________

1. Start stopped and set the checked low speed.

2. Drive forward for 0.8 seconds, then stop and end. Set the collection box using the distance measured in lesson 2.

3. Do not add continuous driving. Return the paper pieces and robot to their marks between tests.

My prediction:

Run	Pieces in box (0-3)	Wheels clear?	Stopped by itself?
1			
2			
3			

Improve your chosen build

What counts as success?

- ☐ At least two of the three tissue squares reach the box.
- ☐ No paper gets caught in the wheels.
- ☐ The robot stops by itself and touches no object except the tissue squares.

Change one thing. What did you change?

Run	Pieces in box (0-3)	Wheels clear?	Stopped by itself?
1			
2			
3			

Did shorter or longer bristles move the paper better?

Which result supports your answer?

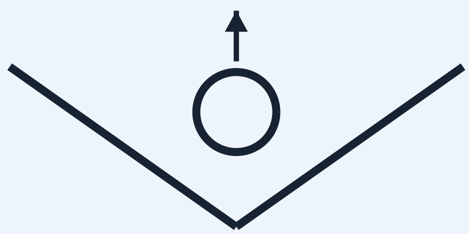
Choice 4: Ball-pushing robot

Name: ______________________________

Date: ____________________ **Period:** ____________________

Guide one light paper ball toward a goal.

LOOK AT THE IDEA



Top view: a shallow V

The opening faces forward.

Leave room for the ball to roll.

Gather your materials

- The shared checked base
- Thin card about 14 by 5 cm
- One crumpled paper ball about 3 cm wide
- Removable tape and ruler

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

Build your attachment

1. Fold the card lengthwise so it has a 3 cm upright wall and a 2 cm mounting flap. Fold it gently across the middle to make a shallow V.
2. Make one light paper ball about 3 cm wide. Put it in the V opening. Leave enough room for it to roll freely; do not make a tight clamp.
3. Turn the robot off. Attach the two mounting flaps to teacher-approved fixed front points. Keep the V open at the front. Leave the wheels clear and keep the card just above the floor.
4. Keep the sensor clear even though this task uses a timed drive. Mark a wide goal across a short straight lane. Place the ball at the same start mark each time.
5. Use the short program below. The robot pushes the ball; it does not grip it. Stop if the ball rolls toward people or outside the clear test area.
6. Repeat three times. If the ball escapes sideways, adjust the V angle a little. Keep the same ball and code while comparing.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Program and first tests

Name: ______________________________

Date: ____________________ **Period:** ____________________

1. Start stopped and use the checked low speed.

2. Drive forward for 0.8 seconds, then stop and end. Put the goal within the tested travel distance.

3. No automatic repeat. Return the robot and ball to their marks with the power off.

My prediction:

____

Run	Ball crossed goal?	Guide stayed on?	Stopped by itself?
1			
2			
3			

Improve your chosen build

What counts as success?

- ☐ The ball crosses the marked goal line.
- ☐ The guide stays attached and the wheels turn freely.
- ☐ The robot stops by itself and touches only the paper ball.

Change one thing. What did you change?

Run	Ball crossed goal?	Guide stayed on?	Stopped by itself?
1			
2			
3			

Which V angle kept the ball in front of the robot?

Which result supports your answer?

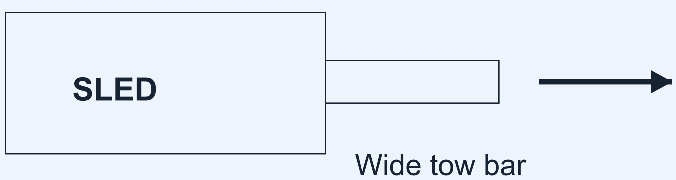
Choice 5: Rescue carrier

Name: ______________________________

Date: ____________________ **Period:** ____________________

Tow a light paper figure on a small rescue sled.

LOOK AT THE IDEA



Attach behind the robot.

Keep away from wheels.

Use a straight route.

Gather your materials

- The shared checked base
- Thin card: one 12 by 6 cm piece and one 12 by 2 cm strip
- A flat paper person, two paper strips for bands, removable tape and scissors
- Ruler and a clear straight lane

Use the same teacher-checked base. The drawing explains the attachment; it is not a full-size cutting template.

Build your attachment

1. Use the 12 by 6 cm card as a sled. Fold each long edge up by 1 cm. Curl the front edge slightly upward so it does not catch on the floor.
2. Draw and cut out a small paper person. Lay it flat on the sled. Add two loose paper bands across the sled and tape their ends to the sides to hold the figure in place.
3. Use the 12 by 2 cm card strip as a wide tow bar. Tape one end flat to the sled's front. Make a small folded tab at the other end.
4. Before attaching it to the robot, pull the sled gently by hand using its wide tow bar. It should slide without catching. Keep it light and use only a straight route for this task.
5. Turn the robot off. Ask the teacher to connect the tow-bar tab to a fixed rear mounting point. Keep the whole strip centered behind the robot and away from the wheels. Do not use loose string.
6. Run the short program below three times. If the sled catches or a wheel struggles, stop. Smooth the sled edge or shorten the route, then test again.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Program and first tests

Name: ____________________

Date: ____________________ **Period:** ____________________

1. Start stopped and use the checked low speed.
2. Drive straight for 0.8 seconds, then stop and end. Do not add turns or reversing for this first sled build.
3. Return the robot and sled to the start with the power off. Do not move a real person, animal or heavy object.

My prediction:

Run	Person stayed on?	Tow bar clear?	Moved and stopped?
1			
2			
3			

Improve your chosen build

What counts as success?

- ☐ The paper person stays on the sled.
- ☐ The tow bar stays attached and away from the wheels.
- ☐ The robot moves the sled and stops by itself.

Change one thing. What did you change?

Run	Person stayed on?	Tow bar clear?	Moved and stopped?
1			
2			
3			

What helped the sled slide without catching?

Which result supports your answer?